

TruthLens AR: Bangla Multimodal Fake News, Image & Deepfake Detection Using Vision-Language Models

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I. INTRODUCTION

The rapid dispersion of deceptive multi-modal digital assets across South Asian social media channels poses an unprecedented threat to public information integrity. While automated, large-scale fact-checking frameworks have matured significantly for dominant, high-resource languages like English, engineering accurate digital verification mechanisms tailored specifically for the Bangla language ecosystem presents critical operational challenges. This domain bottleneck stems from a combination of lexical scarcity, localized colloquial variations, and distinct cultural nuances that traditional text-filtering tools completely fail to capture.

Furthermore, contemporary disinformation operations have evolved beyond isolated textual claims, increasingly utilizing sophisticated visual manipulations, context-swapped graphic layers, and mismatched caption frameworks to bypass standard verification channels. To resolve this text-centric blindspot, this work introduces **TruthLens**, a unified, cross-modal verification paradigm designed to protect users directly at the point of physical media consumption. By routing live video input frames through cloud-hosted Vision Language Models (VLMs), TruthLens synthesizes deep visual features with localized transformer semantic tokenization to decode complex contextual contradictions. This structural assessment is then streamed directly to a mobile frontend utilizing Google's ARCore SDK, which projects stable, color-coded cryptographic verification alerts and explainable reasoning overlays over target printed text or screens in real time.

II. LITERATURE REVIEW

A. Overview

The proliferation of deceptive digital content across social media networks presents a critical challenge to information integrity. While automated fact-checking systems have advanced significantly for high-resource languages like English, research focused on the Bangla language ecosystem remains severely constrained. This disparity creates an operational vulnerability; standard text-filtering tools fail to capture the localized semantic nuances, regional dialects, and cultural metaphors unique to the region. Furthermore, modern disinformation strategies

increasingly exploit cross-modal manipulation, combining altered visual assets with misleading textual context.

To address this vulnerability, this work identifies an intersection between emerging multi-modal deep learning architectures and spatial computing interfaces. Vision Language Models (VLMs) have demonstrated significant capacity for synchronous image-text processing, enabling deeper contextual analysis of mixed-media content than standalone natural language processing (NLP) pipelines. Concurrently, Augmented Reality (AR) frameworks offer a mechanism for real-time, context-aware information visualization. By integrating these domains, **TruthLens** addresses the multi-modal verification gap in Bangla media, processing complex media streams via cloud-hosted VLMs and projecting localized verification overlays directly onto a user's physical environment.

B. Fake News Detection

1) *Dataset-Based Approaches*: The development of supervised machine learning classifiers for misinformation detection relies heavily on the availability of standardized, labeled benchmark datasets. Initial efforts in automated fact-checking were predominantly established using Western political archives. A foundational benchmark is the LIAR dataset [5], which comprises over 12,000 short statements extracted from PolitiFact and structured with a six-class fine-grained truth scale ranging from "true" to "pants-on-fire." While LIAR validated the utility of integrating speaker metadata and historical records into verification pipelines, its structural optimization for English text limits its direct applicability to the linguistic patterns of South Asian digital media.

To mitigate this resource scarcity within the Bangla language ecosystem, researchers have focused on constructing native benchmarks from scratch. Hossain et al. introduced BanFakeNews [1], the first large-scale text corpus for Bangla fake news detection, containing approximately 50,000 crawled articles from mainstream online news portals. This dataset formalized key challenges native to Bangla digital media, specifically

highlighting the high prevalence of structural sarcasm and severe headline-body inconsistencies.

As digital deception migrated from structured news portals to conversational platforms, text-only detection frameworks encountered significant performance drops. Kamruzzaman et al. addressed this shift by developing the BanMANI dataset [2], which focuses explicitly on manipulated conversational text patterns across social media networks. Rather than classifying isolated articles, BanMANI evaluates user comments and posts relative to verified reference articles, shifting the research focus toward tracking active online information distortion.

2) *Overcoming Technical Bottlenecks*: Early computational models trained on native Bangla datasets faced severe constraints due to extreme class imbalance, where authentic samples vastly outnumbered deceptive instances. For example, the original BanFakeNews corpus exhibited a 97% majority class bias, causing standard classifiers to overfit toward predicting authentic labels.

Recent literature focuses extensively on optimizing classification boundaries under highly skewed distributions. Hossain et al. evaluated data-level augmentation frameworks, demonstrating that combining Synthetic Minority Over-sampling Techniques (SMOTE) with meta-model stacking architectures stabilizes prediction boundaries and lowers false-negative rates [4]. This approach proved that ensemble systems combining multiple statistical perspectives outperform standalone classifiers when training metrics are highly uneven.

More recently, the methodology has transitioned from static feature extraction toward generative data augmentation and deep semantic modeling. Shibu et al. developed BanFakeNews-2.0 [3], expanding the initial benchmark by introducing thousands of expert-curated fake samples distributed across 13 distinct contextual categories. By replacing classical statistical extractors with fine-tuned transformer-based architectures and Large Language Models (LLMs), their framework confirmed that deep tokenization models capture localized syntax and cultural nuances far better than classical machine learning algorithms.

TABLE I
STRUCTURAL AND ARCHITECTURAL METRICS OF CORE BENCHMARKS

Dataset	Lang.	Source Type	Size	Primary Limitation
LIAR [5]	English	Political Claims	12.8K	Lacks multi-modal assets
BanFake [1]	Bangla	Online Portals	~50K	Text-only; lacks social context
BanMANI [2]	Bangla	Social Streams	Variable	Passive; no real-time pipeline
BanFake-2.0 [3]	Bangla	Cross-Platform	Expanded	High backend latency loop

C. Methodological Evolution and the Text-Centric Blindspot

1) *Classical ML vs. Transformer Architectures*: The computational methodologies deployed within the Bangla verification domain have undergone a distinct architectural evolution. Initial baseline frameworks relied heavily on hand-crafted linguistic features, TF-IDF vectors, and traditional statistical classifiers such as SVM and Random Forests [1]. These classical models treated words as isolated tokens, which restricted their ability to capture semantic dependency, resulting in poor performance when parsing complex sentence structures or sarcasm.

The integration of model stacking and ensemble learning mitigated these performance drops by feeding the outputs of heterogeneous base models into a unified meta-classifier [4]. However, the primary paradigm shift occurred with the adoption of deep transformer architectures and localized LLMs [3]. Fine-tuned language models replaced rigid keyword matching with contextual tokenization, allowing downstream classifiers to evaluate semantic intent and regional social media slang.

2) *The Multi-Modal Disconnect*: Despite these algorithmic advances, contemporary Bangla fact-checking methodologies share a critical structural vulnerability: they operate with a strict, text-centric focus. Modern disinformation campaigns on platforms like Facebook or WhatsApp rarely rely on text in isolation; instead, they utilize context-swapped imagery, doctored graphics, or mismatched captions.

Because existing regional frameworks analyze textual strings independently of accompanying media assets, they remain vulnerable to visual manipulation. A transformer model may evaluate a text caption as authentic based on its clean semantic signature, completely missing the fact that the associated image has been digitally altered. This text-isolated boundary represents a significant blindspot in current state-of-the-art Bangla verification systems.

D. Vision Language Models (VLMs)

Transitioning from text-isolated systems to multi-modal verification requires models capable of evaluating image-text relationships simultaneously.

1) *Contrastive Image-Text Alignment (CLIP)*: The foundational protocol for joint vision-language representation was established by CLIP [6]. Rather than executing traditional pixel-to-word prediction, CLIP employs a dual-encoder architecture optimized via a contrastive learning objective. This forces matching image-text pairs into close proximity within a shared vector space. It demonstrates robust zero-shot cross-modal retrieval, but operating on a strict contrastive matching bottleneck, CLIP lacks a generative language decoder. It cannot execute open-ended logical reasoning or generate descriptive validation summaries, and it is fundamentally incapable of independent fact verification.

2) *Parameter-Efficient Modality Bridging (BLIP-2)*: To introduce generative reasoning capabilities without incurring the extreme computational cost of training end-to-end multi-modal networks, BLIP-2 introduced a bootstrapped pre-training framework [7]. This architecture utilizes a lightweight, 12-layer Querying Transformer (Q-Former) to bridge a frozen visual backbone with a frozen Large Language Model, serving as an efficiency-focused prompt generator. It achieves high parameter efficiency during training, but the runtime inference latency remains high due to the multi-billion parameter frozen LLM backbone. This creates a computational barrier for deployment on mobile hardware.

3) *Visual Instruction Tuning (LLaVA)*: Representing the transition toward multi-modal conversational agents, LLaVA integrates a visual backbone with a language decoder using a linear projection matrix, optimized through visual instruction tuning [8]. It demonstrates advanced visual reasoning and deep scene composition analysis, allowing the model to infer

intent and identify context-swapped media assets. However, the architecture exhibits a vulnerability to visual hallucination, occasionally manufacturing non-existent details or asserting false data confidently, which risks introducing false positives into verification systems.

E. Spatial Computing and Interface Delivery

1) *Foundational Taxonomy of Mixed Reality*: To effectively deliver verification metrics to a user during active media consumption, the system interface must move past traditional desktop dashboards. The theoretical basis for this spatial transition is grounded in the Reality-Virtuality (RV) Continuum defined by Milgram and Kishino [10]. This taxonomy maps the integration of physical and digital environments along a linear spectrum, isolating Augmented Reality (AR) near the physical anchor where a user’s direct view of the real world is augmented by computer-generated graphical assets. Maintaining an accurate spatial understanding of the physical coordinates is essential to ensure that data overlays remain fixed and do not disorient the user.

2) *Environmental Mapping via ARCore Architecture*: Implementing this spatial continuum on standard mobile hardware requires specialized software development kits that do not rely on external depth sensors. Google’s ARCore SDK [9] achieves stable spatial tracking by running three concurrent mapping pipelines: 1) Concurrent Odometry and Mapping (SLAM) to pair visual feature points from the camera feed with inertial data from the device’s IMU, calculating the smartphone’s exact 3D position over time; 2) Environmental Understanding to group clusters of feature points sharing common mathematical depths to define plane geometries; and 3) Light Estimation to analyze real-time pixel data to estimate ambient lighting intensity, matching the digital assets to physical environmental lighting.

By compiling these outputs into a coordinate transform matrix, ARCore binds digital data to persistent physical coordinates via localized anchors, ensuring the graphic interface tracks accurately with changes in the user’s camera perspective.

3) *The Spatial Visualization Gap*: A critical gap exists in the application domain of these spatial tracking frameworks. Current AR systems are heavily isolated within commercial navigation, interactive education, and mobile gaming [9]. Conversely, existing fact-checking pipelines remain entirely passive, operating on a pull-based model where users must manually extract claims, navigate to external databases, and interpret text-heavy verdicts. No current architecture integrates native Bangla transformer models, deep vision-language verification, and mobile spatial tracking into a unified interface.

TruthLens directly addresses this gap by re-purposing AR tracking as an active cognitive defense layer. Instead of generating gaming assets, the ARCore frontend captures physical media streams and projects color-coded verification indicators and VLM-generated reasoning summaries directly over the source material in real time.

F. Identified Research Gaps & TruthLens Alignment

- 1) **The Language-Modality Divide**: High-performance VLMs are predominantly optimized for English datasets,

while state-of-the-art Bangla fact-checking systems are attuned to local context but remain entirely restricted to text-only pipelines.

- 2) **The Verification Blindspot in VLMs**: Advanced multi-modal models evaluate descriptive correspondence (whether a text matches an image) rather than verifying the objective truth of the underlying claim.
- 3) **The Interface Delivery Bottleneck**: Fact-checking platforms rely on high-friction manual lookups via web dashboards, while high-speed spatial rendering tools like ARCore remain unutilized for data validation.

TruthLens resolves these structural gaps by establishing an integrated, multi-modal verification loop that routes real-time video frames through a cloud-hosted Vision Language Agent and projects immediate, context-aware visual overlays back onto the physical source asset. By linking localized Bangla transformer logic with the multi-modal reasoning of a VLM agent, the system neutralizes the text-isolated blindspot. It then deploys the motion tracking and light estimation of ARCore to transform the smartphone display into an immediate, real-time cognitive barrier against digital misinformation.

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